

SnapVid

Offline Agentic AI Video Maker · On-Device Snapdragon NPU

Turning a plain-text prompt into a polished, narrated, animated video — **in under 60 seconds** — with **zero cloud, zero subscription, zero internet.**

github.com/GoDxVictoryRR/snapvid

MIT Open Source

74/74 Tests Passing

The Problem

Professional explainer videos are expensive, cloud-locked, and privacy-invasive.

■ **5,000**

Minimum cost per HeyGen / Synthesia
video

0 Mbps

Internet available in rural Indian classrooms

100%

Of these tools upload your private content to the cloud

■ **Educators, coaches, and small businesses** need video content but cannot afford cloud video tools.

■ **Privacy matters** — lecture notes, trade secrets, and student data should **never** leave the device.

■ **Snapdragon X Elite laptops** have the NPU horsepower — but no tool uses it for video creation.

■ **Bandwidth is scarce** — offline-first tools are essential for Tier-2/3 cities and remote areas.

The Solution

SnapVid — One prompt. 60 seconds. One polished video. No cloud.

■■ Browser UI

Dark-mode studio with live agent activity log, settings panel, history, and built-in video player.

■■ CLI

`python run.py "topic"` — headless, scriptable, batch-ready.

■ Python API

`from snapreel.planner import generate` — importable into any application.

How it works:

- User types a topic in plain English
- Scene Planner Agent generates structured JSON scene script via NPU
- Self-healing validator checks and repairs the JSON (3 retries)
- Quality Critic Agent reviews pacing and narration
- Kokoro TTS synthesises audio offline — word-level caption sync
- Pure-Python renderer draws animated frames at 30 FPS
- FFmpeg muxes video + audio → polished MP4 in ~30 seconds

Technical Architecture

Two-agent pipeline · Pure Python · Zero Node.js · Zero cloud runtime dependencies

Component	Technology	Detail
LLM Inference (NPU)	Qualcomm GenieX · Hexagon NPU · w4a16	OpenAI-compatible endpoint · auto-selected by provider setting
Agent Loop	Scene Planner + Quality Critic · Pydantic v2	JSON validation · self-healing repair prompt · SSE event stream
Frame Renderer	Pillow ≥ 10 · NumPy ≥ 1.26 · 1280×720 @ 30 FPS	10 animated templates · easing curves · grain overlay
Transitions	Pure-Python compositor	Fade · slide · zoom-in · cross-dissolve between scenes
TTS Engine	Kokoro 82M → Edge TTS → SAPI	Auto-selected · word-level caption timing · fully offline
Video Mux	imageio-ffmpeg (bundled)	No system ffmpeg install · H.264 + AAC MP4 container
Web UI	FastAPI · SSE · HTML/CSS/JS	Real-time agent log · settings modal · history · video player
Tests	pytest · 74 tests · 100% passing	Schema · LLM · renderer · TTS · server · transitions · NPU · history

Snapdragon NPU Performance

Verified on Snapdragon X Elite CRD via Qualcomm AI Hub · GenieX w4a16 quantization

~22.4

Tokens / Second on Hexagon
NPU

~30s

Full 60-second video,
end-to-end

2x

Faster than Real-Time on Snapdragon X Elite

Pipeline Stage	Hardware	Latency	Throughput
Qwen3-4B Scene Planner	Hexagon NPU (GenieX w4a16)	~2.1 s / script	~22.4 tok/s
Kokoro TTS	CPU ARM64	~1.8 s / scene	Real-time ×1.4
Pillow Renderer	CPU ARM64	~0.6 s / scene	30 FPS native
Full Pipeline	Snapdragon X Elite	~30 s total	60 s video

Competitive Landscape

SnapVid wins on every axis that matters for offline, privacy-first video creation.

Feature	SnapVid	HeyGen / Synthesia	Manim / Remotion
Offline / No internet	■ Full NPU pipeline	■ Cloud required	■ ■ Partial
Cost	Free · MIT Open Source	\$24–\$119 / month	Free (expert setup)
Privacy (data stays on device)	■ Never leaves device	■ Uploads content	■ Local
Setup time	pip install + 1 command	Account + subscription + onboard	Hours (LaTeX / Node deps)
60 s video speed	~30 s on Snapdragon	Minutes (cloud queue)	■ Render compile hours
Snapdragon NPU acceleration	■ Hexagon NPU via GenieX	■ None	■ None
API / CLI access	■ Browser + CLI + Python API	Web only	Script only

Key Innovations

Six technical breakthroughs that make SnapVid uniquely powerful.

■ Self-Healing JSON Agent

Invalid LLM output triggers automatic repair prompt loop — up to 3 retries — before failing gracefully.

■ Audio-First Rendering

Narration synthesised first; scene durations reconciled to audio length — guaranteeing perfect A/V sync.

■ Word-Level Caption Sync

On-screen captions timed to individual word offsets within each audio chunk — karaoke-style display.

■ Two-Agent Architecture

Scene Planner agent generates content; Quality Critic agent independently reviews pacing, clarity and narration before render.

■ Animated Transitions

Smooth fade, slide, zoom-in, and cross-dissolve transitions composited in pure Python — no external video library.

■ Live-Config from UI

LLM provider, model, API key, voice, and format all switchable live from the settings drawer — no code changes.

Target Users & Real-World Impact

SnapVid democratizes professional video creation for the next billion users.

■ School & Coaching Teachers

Generate curriculum explainer videos for offline classrooms. Zero cost. Zero internet.

■ YouTube Educators

Rapid STEM tutorial video drafts without a recording studio or video editor subscription.

■ Corporate L&D; Teams

Internal training modules with zero data-exposure risk — content stays on-device.

■ Small Business Owners

Product explainers in local language, offline, at zero cost — built in under 60 seconds.

■ Developers & Data Scientists

Automated documentation, concept visualization pipelines, and API-first video generation.

■ NGOs & Rural Education

Offline video content creation for areas with no reliable internet — runs on battery power.

Technology Stack & Demo

Fully open source · MIT License · github.com/GoDxVictoryRR/snapvid

Core Stack

- **Python 3.11+** — Pillow, NumPy, FastAPI, Pydantic v2
- **imageio-ffmpeg** — Bundled FFmpeg binary
- **Qualcomm GenieX** — NPU LLM inference
- **Kokoro 82M TTS** — Offline neural voice
- **Ollama / llama.cpp** — CPU/GPU fallback
- **OpenAI / Groq / NIM / Gemini** — Cloud options

Supported Providers

- Qualcomm GenieX (NPU) · Ollama · llama.cpp
- OpenAI · Groq · NVIDIA NIM · Google Gemini
- Together AI · Mistral AI · Any OpenAI-compatible API

Quick Start

```
git clone github.com/GoDxVictoryRR/snapvid cd snapvid pip install -r requirements.txt cp .env.example .env python -m snapreel.server
```

Test Suite

74 / 74 tests passing

Schema · LLM · Renderer · TTS · Server · Transitions · NPU · History

SnapVid

The only offline, NPU-accelerated, privacy-first AI video maker.

■ Free Forever

■ 30 sec / video

■ 100% On-Device

■ MIT Open Source

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